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CS3807 – Deep Learning Laboratory

Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

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Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27
Platform	Google Colab (GPU), Python 3.13, TensorFlow/Keras 2.21.0, TensorFlow Datasets 4.9.10, seed 42	

1. Aim

The aim of this experiment is to study how different attention mechanisms modify a pretrained ResNet50 (ImageNet weights) and how the **position at which attention is inserted** influences image-classification performance on the five-class `tf_flowers` dataset. The experiment compares channel, spatial, sequential channel–spatial and lightweight self-attention modules; inserts the same module after conv3_x, conv4_x and conv5_x (P_2, P_3, P_4) and at P_3+P_4; examines the attention order and the spatial-kernel size; compares the results with ECA and CBAM; and interprets the models with accuracy, macro precision/recall/F1, parameters and inference time (quantitative and complexity analysis) and with feature maps, internal attention maps and Grad-CAM-style overlays (qualitative analysis). All comparisons use the same backbone, data split and training protocol so that only the studied factor changes.

2. Dataset Description

Recommended dataset: TensorFlow Flowers (`tf_flowers`). It contains natural RGB flower images of five classes (daisy, dandelion, roses, sunflowers, tulips) with variation in scale, viewpoint, illumination, background clutter, flower size, colour, texture and object location, which makes it suitable for studying attention. Every image is resized to $224 \times 224 \times 3$ and preprocessed with the ImageNet ResNet50 procedure; the split is 70% : 15% : 15% (training : validation : testing) and the same samples are kept in each subset for all experiments.

The `tf_flowers` dataset was loaded through TensorFlow Datasets and contains **3,670** RGB images. TFDS indexes the classes as `0=dandelion, 1=daisy, 2=tulips, 3=sunflowers, 4=roses`; this order is used in every table and confusion matrix below. The split was drawn *per class* (stratified) with a fixed generator (`numpy.random.default_rng(42)`) using $\lfloor 0.70n \rfloor$ training, $\lfloor 0.15n \rfloor$ validation and the remaining images for testing, and the three index arrays were then reused unchanged for all ten models.

Table 1: Class distribution of the fixed 70:15:15 split (actual proportions 69.9 % : 14.9 % : 15.1 %).

Class	Train	Validation	Test	Total
dandelion	628	134	136	898
daisy	443	94	96	633
tulips	559	119	121	799
sunflowers	489	104	106	699
roses	448	96	97	641
Total	**2567**	**547**	**556**	**3670**

Preprocessing. Every image is resized to $224 \times 224 \times 3$ and, for the *training set only*, passed through the augmentation stack `RandomFlip(horizontal)`, `RandomRotation(0.05)` ($\approx \pm 18°$), `RandomZoom(0.10)`, `RandomTranslation(0.05,0.05)` and `RandomContrast(0.10)`; afterwards `resnet50.preprocess_input` is applied and labels are one-hot encoded. The batch size is 32, giving image batches of shape $(32, 224, 224, 3)$ and label batches of shape $(32, 5)$.



Figure 1: One sample image from each of the five classes (dandelion, daisy, tulips, sunflowers, roses).

Inference. The classes are reasonably balanced (largest: dandelion, 24.5 %; smallest: daisy, 17.2 %), so accuracy and macro-averaged metrics stay close to each other and no class weighting was needed. The samples show the intended variation in scale, viewpoint, illumination, background clutter and colour (e.g. the dandelion on a dark background, the daisy with an insect, the partly grey-scale roses). **Resolution of the experiment:** the test set has only 556 images, so one image equals 0.18 percentage points (pp), and the binomial standard error of an accuracy near 92 % is $\sqrt{0.92 \cdot 0.08/556} \approx 1.15$ pp. Differences between models of roughly 1–1.5 pp or less are therefore only suggestive, particularly because each configuration was trained once (seed 42).

3. ResNet50 Architecture

ResNet50 pretrained on ImageNet is the only backbone. For an input $I \in \mathbb{R}^{224\times224\times3}$ its stages produce the following feature tensors; the candidate attention positions are the stage outputs.

Table 2: ResNet50 stages and the candidate attention positions.

Stage	Position	Resolution	Channels	Typical content
Stem + MaxPool	–	56×56	64	edges, colours
conv2_x	P_1	56×56	256	edges, orientations, simple textures
conv3_x	P_2	28×28	512	local patterns, petal arrangements
conv4_x	P_3	14×14	1024	flower parts, shapes
conv5_x	P_4	7×7	2048	class-discriminative semantics

With depth the spatial resolution decreases while the number of channels grows, and the representation moves from low-level to high-level information. The same attention module can therefore behave differently at different depths: early attention acts on high-resolution, low-level features, middle attention on mixed spatial/semantic features, and late attention on low-resolution semantic features. P_1 (56×56) was not run because its high resolution makes attention costly and the manual recommends concentrating on P_2–P_4.

Baseline model. The baseline is the pretrained ResNet50 without attention, followed by Global Average Pooling, Dropout and a 5-class Dense layer.



The baseline is `ResNet50(include_top=False, weights='imagenet')` followed by Global Average Pooling, Dropout (0.3) and `Dense(5, softmax)`. It has **23,597,957** parameters in total (23,587,712 in the backbone and $2048 \cdot 5 + 5 = 10{,}245$ in the classifier). In phase 1 only the classifier head is trainable; in phase 2 the convolution layers from `conv5_block1_1_conv` onwards are also unfrozen (BatchNorm layers remain frozen), giving **14,963,717** trainable parameters. All attention variants are obtained from this model by rebuilding the ResNet50 graph and inserting the module on the output tensor of the chosen stage (`conv2_block3_out`, `conv3_block4_out`, `conv4_block6_out` or `conv5_block3_out`).

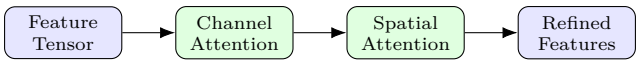
4. Attention Mechanisms

Let $F \in \mathbb{R}^{H\times W\times C}$ be the feature tensor at the insertion point.

Channel attention (also SE). Global average pooling gives $z_c = \frac{1}{HW}\sum_{i,j} F(i,j,c)$, and $M_c(F) = \sigma(W_2\,\delta(W_1 z))$, $F_c = F \odot M_c(F)$, with reduction ratio $r = 16$. It answers "which channels are important?".

Spatial attention. $F_{\text{avg}} = \text{Mean}_c(F)$, $F_{\max} = \text{Max}_c(F)$, $M_s(F) = \sigma(\text{Conv}_{k\times k}([F_{\text{avg}} \| F_{\max}])) \in \mathbb{R}^{H\times W\times 1}$ and $F'_s = F \odot M_s(F)$ ($k = 7$ by default). It answers "which locations are important?".

Sequential channel–spatial attention. $F_c = M_c(F) \odot F$ and $F_{cs} = M_s(F_c) \odot F_c$ (what, then where); the reverse order is Spatial→Channel.



Self-attention. Flatten to $X \in \mathbb{R}^{N \times C}$ with $N = HW$ and compute $Q = XW_Q$, $K = XW_K$, $V = XW_V$ (key dimension $d = 128$), $A = \mathrm{softmax}(QK^T/\sqrt{d})$, $Y = AV$ and $F_{\mathrm{SA}} = F + \gamma\,\mathrm{reshape}(Y)$ with a learnable γ initialised to 0 (the block starts as an identity). It answers "which locations are related to which?". The $N \times N$ matrix grows quadratically with the resolution ($N = 196$ at P_3, $N = 49$ at P_4), so self-attention was used at P_3.

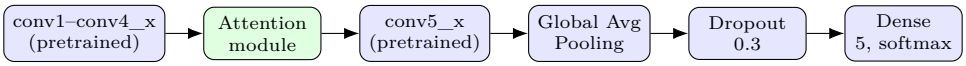
ECA, CBAM and the combined module. ECA replaces the bottleneck MLP of channel attention by a 1-D convolution ($k = 3$) over the pooled channel descriptor (3 parameters). CBAM applies a channel branch (shared MLP on both average- and max-pooled descriptors, outputs summed before the sigmoid) followed by the 7×7 spatial branch. The "Combined" model applies Channel→Spatial and then self-attention.

5. Proposed Experimental Architecture

For a selected position P_k the network is

$$\text{Previous ResNet stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet stage},$$

for example (attention at P_3):



The attention module is implemented by rebuilding the ResNet50 graph and inserting the layer on the output tensor of the chosen stage; it does not change the feature dimensions, as the table below shows.

Table 3: Feature-tensor shape (H, W, C) immediately before and after each attention module (batch dimension omitted).

Model	Position	Input	Output	Added parameters
Channel (SE, r=16)	P_3	$(14, 14, 1024)$	$(14, 14, 1024)$	132,160
Spatial (7×7)	P_3	$(14, 14, 1024)$	$(14, 14, 1024)$	98
Channel→Spatial	P_2	$(28, 28, 512)$	$(28, 28, 512)$	33,410
Channel→Spatial	P_3	$(14, 14, 1024)$	$(14, 14, 1024)$	132,258
Channel→Spatial	P_4	$(7, 7, 2048)$	$(7, 7, 2048)$	526,562
Self-attention (d=128)	P_3	$(14, 14, 1024)$	$(14, 14, 1024)$	524,289
Combined (C+S+SA)	P_3	$(14, 14, 1024)$	$(14, 14, 1024)$	656,547
ECA (k=3)	P_3	$(14, 14, 1024)$	$(14, 14, 1024)$	3
CBAM	P_3	$(14, 14, 1024)$	$(14, 14, 1024)$	132,258

For `P3_channel_spatial_attention` the notebook printed "before $(14, 14, 1024)$ → after $(14, 14, 1024)$". The other rows follow from the stage shapes defined in the code and from the fact that every module returns $F \odot M$ (channel, spatial, ECA, CBAM) or $F + \gamma\,\mathrm{reshape}(Y)$ (self-attention), both of which preserve $H \times W \times C$.

Inference. No module changes the spatial or channel dimensions, so each can be inserted without altering the following ResNet stage. The added parameters follow directly from the formulas: channel attention at P_3 has $(1024 \cdot 64 + 64) + (64 \cdot 1024 + 1024) = 132{,}160$ weights, the 7×7 spatial convolution on the two pooled maps has $2 \cdot 49 = 98$, ECA has the three taps of its 1-D kernel, and the self-attention block has three $1024 \rightarrow 128$ projections, one $128 \rightarrow 1024$ output projection and the scalar γ ($3 \cdot 131{,}072 + 131{,}072 + 1 = 524{,}289$).

6. Experimental Setup

Transfer learning. ImageNet ResNet50 weights are loaded, the classification layer is removed, the attention module, Global Average Pooling, Dropout and a 5-neuron Dense layer are added, the backbone is first frozen and then the final stage is fine-tuned with a smaller learning rate.

Table 4: Training protocol, identical for all ten models.

	Phase 1 (frozen backbone)	**Phase 2 (fine-tuning)**
Epochs	3	2
Optimiser / learning rate	Adam, 10^{-3}	Adam, 10^{-5} (re-compiled)
Trainable layers	attention module(s), Dropout, Dense	as phase 1 + all conv5_x convolutions (BatchNorm frozen)
Batch size / dropout	32 / 0.3	32 / 0.3
Loss / seed	categorical cross-entropy / 42	categorical cross-entropy / 42

A pretrained ResNet50 (ImageNet) with the original classification layer removed is used in every case. The notebook saves weights and history after every epoch, so an interrupted run resumes from its last finished epoch (Adam moment estimates are not stored, so a resumed epoch can differ slightly). **Limitation:** five epochs in total is a short schedule; the training curves are still rising at epoch 5.

Loss. Categorical cross-entropy $\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log\widehat{y}_{i,c}$ is used for every model (one-hot labels, softmax output).

Metrics. Accuracy and macro-averaged precision, recall and F1 are computed from the test confusion matrix: $\text{Precision}_c = \frac{TP_c}{TP_c+FP_c}$, $\text{Recall}_c = \frac{TP_c}{TP_c+FN_c}$, $F1_c = \frac{2P_cR_c}{P_c+R_c}$, Macro F1 $= \frac{1}{C}\sum_{c=1}^{C} F1_c$ ($C = 5$). The number of trainable and total parameters and the mean inference time per image are also recorded.

Experimental protocol. All ablations use identical training, validation and test images, seed (42), preprocessing, augmentation, backbone, classifier, optimiser, batch size, epochs, learning-rate schedule and fine-tuning procedure; only the factor under study changes. The platform was Google Colab (GPU) with Python 3.13, TensorFlow/Keras 2.21.0 and TensorFlow Datasets 4.9.10.

7. Attention-Type Ablation

The first experiment fixes the position at P_3 and changes only the attention type.

Table 5: Design of the attention-type ablation (all at P_3).

Model	**Channel**	**Spatial**	**Self-Attention**
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

Table 6: Test-set results (556 images), attention fixed at P_3. Metrics are macro-averaged percentages; Δ is the accuracy change relative to A0 in percentage points.

Model	**Acc.**	**Δ**	**Prec.**	**Rec.**	**F1**	**Added params**
A0: Baseline ResNet50	92.09	–	92.09	92.08	92.04	0
A1: Channel Attention	92.27	+0.18	92.52	92.10	92.20	132,160
A2: Spatial Attention	91.55	-0.54	91.56	91.42	91.38	98
A3: Channel+Spatial	91.37	-0.72	91.35	91.34	91.25	132,258
A4: Self-Attention	93.53	+1.44	93.56	93.45	93.46	524,289
A5: Combined Attention	93.71	+1.62	93.86	93.59	93.65	656,547

Inference.

- Only the two models that contain self-attention improved clearly: A4 (93.53 %, 520/556 correct) and A5 (93.71 %, 521/556) exceed the baseline (92.09 %, 512/556) by +1.44 and +1.62 pp. Adding channel and spatial attention to self-attention (A5 vs. A4) changed the result by a single image (+0.18 pp), so it brought no measurable benefit.
- Channel attention (A1) is practically identical to the baseline (+0.18 pp, one image), while spatial attention (A2, −0.54 pp) and channel→spatial attention (A3, −0.72 pp) are slightly worse. All three differences are smaller

than one standard error (≈ 1.15 pp), so at P_3 the convolutional gates should be regarded as neither helping nor hurting.

- A plausible explanation (suggested by the design, not isolated by a dedicated ablation) is how the modules start. Channel and spatial attention are *multiplicative sigmoid gates*: at initialisation every weight is ≈ 0.5, so the pretrained conv4_x features are rescaled before they enter the pretrained conv5_x stage, and only five epochs are available to compensate. Self-attention is *residual* with γ initialised to 0, so it starts as an identity mapping, preserves the pretrained representation and can add information gradually.
- Gains are not tied to cost: A1 and A3 add 132k parameters (0.56 %) without benefit, while A4 adds 524k (2.2 %) and gains 1.4 pp; the two models with almost no extra parameters (Spatial: 98; ECA: 3) do not help (Section 16).

Table 7: Attention order at P_3.

Order	**Accuracy**	**Macro F1**	**Total params**	**ms/img**
Channel→Spatial	91.37	91.25	23,730,215	7.53
Spatial→Channel	n/r	n/r	(23,730,215)	n/r

Inference. The notebook contains the code for the Spatial→Channel run (experiment key `order_Spatial_Channel`) but no result for it is stored in the saved notebook outputs (n/r = not recorded), so the two orders cannot be compared empirically here. Both orders contain the same layers, hence the same number of parameters. The Channel→Spatial reference (identical to model A3) is 91.37 %, i.e. not better than channel attention alone. To complete this experiment, re-run the order cell in the notebook and enter the second row.

Spatial-kernel size. Spatial attention at P_3 was planned for $k \in \{3, 5, 7\}$.

Table 8: Spatial-kernel ablation (n/r: run implemented in the notebook but result not stored in the saved outputs).

Spatial Kernel	**Accuracy**	**Macro F1**	**Inference Time**
3×3	n/r	n/r	n/r
5×5	n/r	n/r	n/r
7×7	91.55 %	91.38 %	7.15 ms

Spatial attention at P_3 with $k = 7$ is the model "Spatial P3" of Experiment 1 and its numbers are entered above. The runs for $k = 3$ and $k = 5$ are implemented in the notebook (`spatial_kernel_3`, `spatial_kernel_5`) but their results were not stored in the saved outputs (n/r), so they are not reported.

Inference. For 7×7 the spatial module has just 98 weights and gives 91.55 % (0.54 pp below the baseline). On the 14×14 map of P_3 a 7×7 kernel already covers half of the map width. The attention map it produced (Figure 9, fourth panel) is smooth and almost flat; a kernel-size effect cannot be established without the 3×3 and 5×5 results.

8. Attention-Position Ablation

The second experiment fixes the module as Channel→Spatial and changes only the position; a third experiment places attention at more than one stage.

Table 9: Same Channel→Spatial module inserted at different stages (baseline accuracy: 92.09 %).

Exp.	**Position**	**Acc.**	**Prec.**	**Rec.**	**F1**	**Added params**
B1	After conv2_x (P_1)	not run (optional; see Section 7)				
B2	After conv3_x (P_2)	91.55	91.45	91.45	91.43	33,410
B3	After conv4_x (P_3)	91.37	91.35	91.34	91.25	132,258
B4	After conv5_x (P_4)	93.35	93.41	93.34	93.34	526,562

Inference. The late position is clearly the best: at P_4 the module reaches 93.35 % accuracy (+1.26 pp over the baseline, +1.80 pp over P_2, +1.98 pp over P_3), whereas P_2 (91.55 %) and P_3 (91.37 %) are below the baseline and differ from each other by one image. So the effect of position is not monotone in depth: it is a step between P_3 and P_4. Two properties of P_4 are consistent with this. (i) P_4 is the only position whose output goes straight into the newly trained Global-Average-Pooling and Dense layers; at P_2 and P_3 the re-scaled features feed *pretrained* stages that expect un-gated statistics. (ii) At P_4 the 2048 channels are high-level, class-related detectors, so re-weighting them directly changes the evidence seen by the classifier. The price is parameters: the channel MLP grows with C^2/r, so the module costs 33k parameters at P_2, 132k at P_3 and 527k at P_4 (+2.23 % of the model). Since only one

run per position was made, the P_4 advantage (≈10–11 images) is evidence, but not proof, of a position effect. P_1 (B1) was not run because the manual recommends concentrating on P_2–P_4.

Table 10: Single- versus multi-stage Channel→Spatial attention.

Configuration	**Acc.**	**Prec.**	**Rec.**	**F1**	**Trainable**	**Total**	**ms/img**
Attention only at P_3	91.37	91.35	91.34	91.25	15,095,975	23,730,215	7.53
Attention only at P_4	93.35	93.41	93.34	93.34	15,490,279	24,124,519	7.75
Attention at $P_3 + P_4$	91.19	91.29	91.16	91.05	15,622,537	24,256,777	7.99
Attention at $P_2 + P_3 + P_4$	not run (`RUN_P2P3P4 = False` in the notebook)						

Inference. Using attention at more locations was *not* more effective. $P_3 + P_4$ (91.19 %) is 2.16 pp (≈12 images) *worse* than P_4 alone (93.35 %) although it carries 658,820 extra parameters, and it is no better than P_3 alone (91.37 %). The P_3 module appears to cancel the benefit of the P_4 module: whatever P_3 does to the features, P_4 cannot repair, and the cost grows (parameters +132k over P_4-only, training and inference work for two modules). The answer to the question posed in Section 8 is therefore that more attention is not necessarily more effective.

9. Performance Results

Table 11: Main result table (test set, 556 images).

Model	**Position**	**Accuracy**	**Precision**	**Recall**	**F1**	**Parameters**	**Inference Time**
Baseline	–	92.09	92.09	92.08	92.04	23,597,957	10.04 ms
Channel Attention	P_3	92.27	92.52	92.10	92.20	23,730,117	7.47 ms
Spatial Attention	P_3	91.55	91.56	91.42	91.38	23,598,055	7.15 ms
Channel+Spatial	P_2	91.55	91.45	91.45	91.43	23,631,367	9.18 ms
Channel+Spatial	P_3	91.37	91.35	91.34	91.25	23,730,215	7.53 ms
Channel+Spatial	P_4	93.35	93.41	93.34	93.34	24,124,519	7.75 ms
Self-Attention	P_3	93.53	93.56	93.45	93.46	24,122,246	10.33 ms
Combined (C+S+SA)	P_3	93.71	93.86	93.59	93.65	24,254,504	10.16 ms
Additional-1: ECA	P_3	91.19	91.05	91.14	91.08	23,597,960	9.48 ms
Additional-2: CBAM	P_3	91.73	92.00	91.40	91.60	23,730,215	10.56 ms

All values are macro-averaged test-set percentages for the same 556 test images; "Parameters" is the total parameter count and "Inference Time" the notebook's mean time per image (see Section 10 for its reliability). The additional mechanisms are ECA (Additional-1) and CBAM (Additional-2); the SE mechanism is identical to "Channel Attention". The combined model (C+S+SA, 93.71 % accuracy, 93.65 % F1, 24,254,504 parameters) is reported in Section 7.

Inference. The ten models fall into three groups: (i) the three models with a large late or residual module – Self-Attention (P_3), Channel+Spatial (P_4) and the Combined model – at 93.3–93.7 % accuracy; (ii) models statistically indistinguishable from the baseline (91.4–92.3 %): Channel, Spatial, Channel+Spatial at P_2 and P_3, and CBAM; and (iii) ECA, the lowest at 91.19 %. Precision, recall and F1 move together with accuracy (for each model all three lie within 0.5 pp of the accuracy), so the ranking does not depend on the chosen metric.

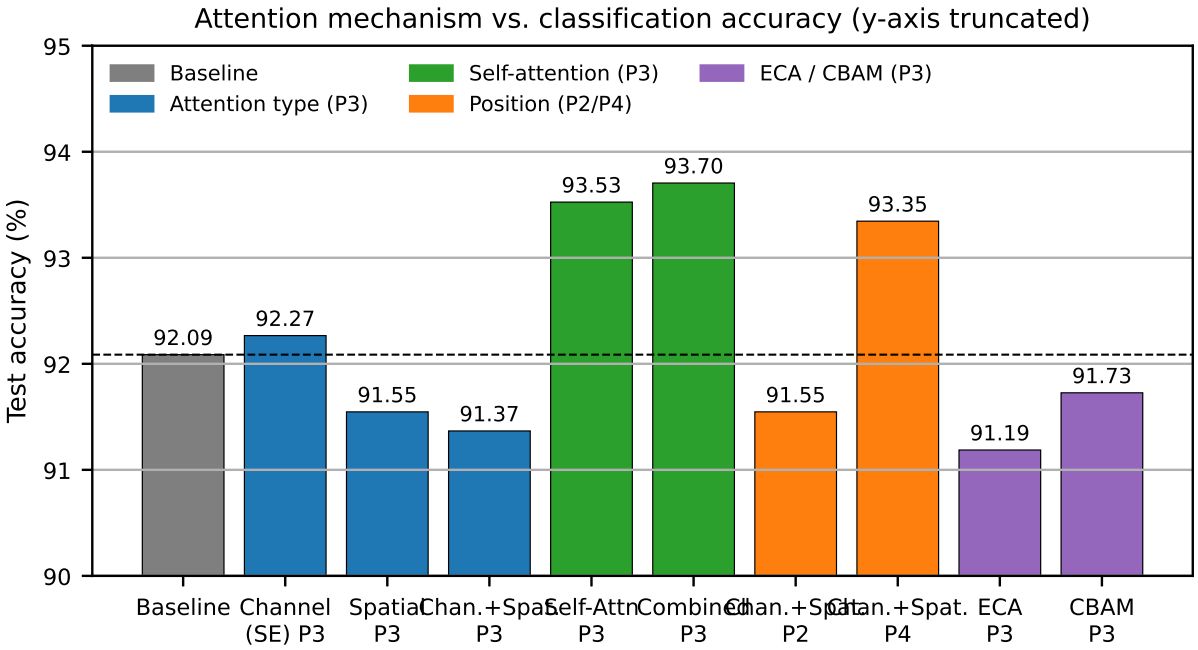


Figure 2: Test accuracy of every model (dashed line: baseline). The y-axis starts at 90 % to make the small differences visible.

Inference. Four of the nine attention models are above the baseline (Combined, Self-Attention, Channel→Spatial at P_4, and Channel by only +0.18 pp) and five are below it (CBAM, Spatial, Channel→Spatial at P_2 and P_3, ECA). The three models above 93 % stand out by 1.3–1.6 pp; everything else is within ±1 pp of the baseline.

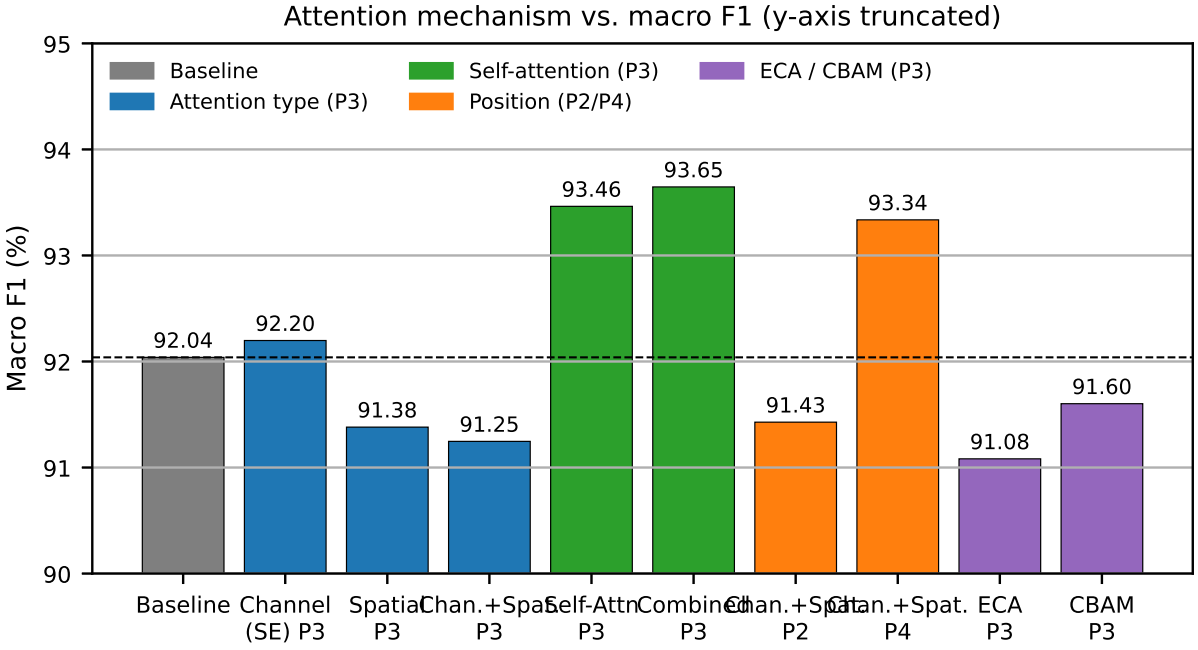


Figure 3: Macro F1-score of every model (dashed line: baseline); y-axis truncated at 90 %.

Inference. The macro-F1 ranking is almost identical to the accuracy ranking (only Spatial P_3 and Channel→Spatial P_2, which tie in accuracy at 91.55 %, swap places). Because the class distribution is fairly balanced, F1 is 0.05–0.2 pp below accuracy for every model, so F1 provides no extra discrimination here; its value is that it would expose a model that gains accuracy by neglecting one class (not observed).

10. Complexity Analysis

Table 12: Parameters (trainable at the end of fine-tuning, and total) and mean inference time per image.

Model	Trainable	Total	Added	Added (%)	ms/img
Baseline	14,963,717	23,597,957	0	0.000	10.04
Channel P3	15,095,877	23,730,117	132,160	0.560	7.47
Spatial P3	14,963,815	23,598,055	98	0.000	7.15
Channel+Spatial P3	15,095,975	23,730,215	132,258	0.560	7.53
Self-Attention P3	15,488,006	24,122,246	524,289	2.222	10.33
Combined P3	15,620,264	24,254,504	656,547	2.782	10.16
Channel+Spatial P2	14,997,127	23,631,367	33,410	0.142	9.18
Channel+Spatial P4	15,490,279	24,124,519	526,562	2.231	7.75
ECA P3	14,963,720	23,597,960	3	0.000	9.48
CBAM P3	15,095,975	23,730,215	132,258	0.560	10.56

How the time was measured. For the first 10 test batches (32 images each) the model is called once as a warm-up and once timed with `time.perf_counter()`, including the transfer of the result to the host; the time divided by the batch size is averaged over the 10 batches. **Inference.** The parameter counts are exact, but the timings should be read with care. A module can only add work, yet the measured times of the baseline (10.04 ms) and ECA (9.48 ms) exceed those of Channel/SE (7.48 ms), Spatial (7.15 ms) and Channel→Spatial at P_3 (7.53 ms), while CBAM, which has the same size and almost the same operations as Channel→Spatial, is 40 % slower (10.56 ms). The 3 ms spread therefore reflects measurement noise (one timed pass per batch on a shared Colab GPU, no repeated runs, no explicit synchronisation), not the modules. Analytically, the heaviest module (self-attention at P_3: ≈0.11 GMAC from the Q/K/V/output projections and the 196×196 attention matrix) is about 2.7 % of ResNet50's ≈4.1 GMAC, and the channel, spatial and ECA modules are well below 1 %.

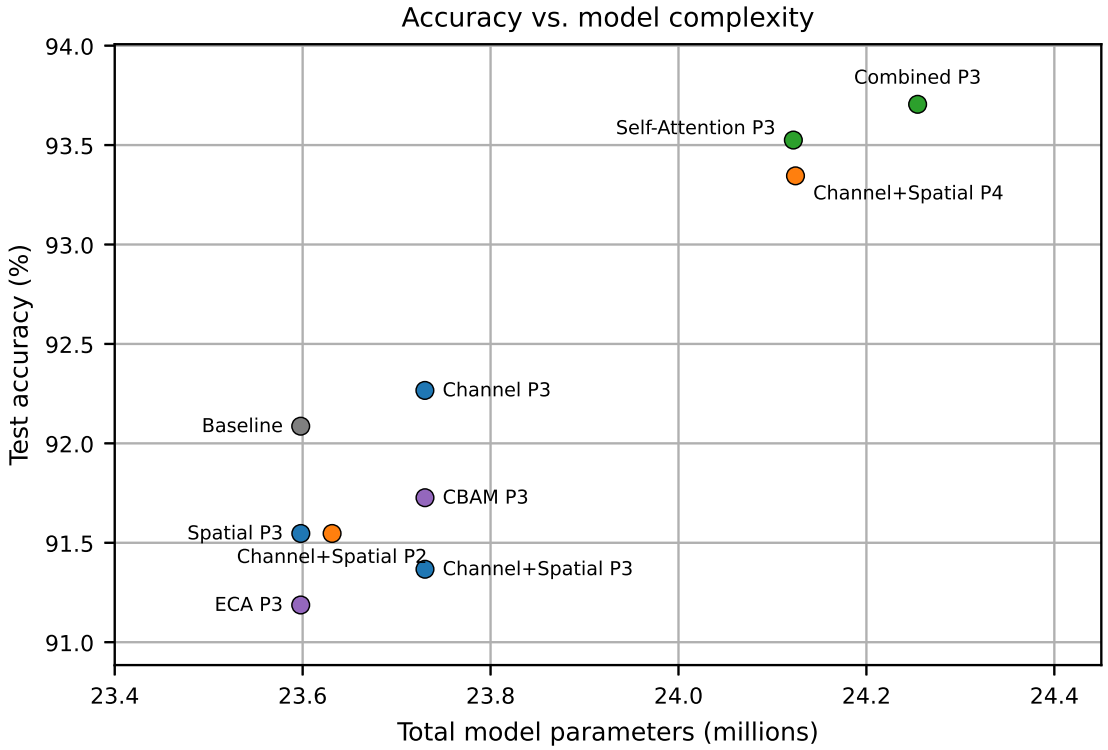


Figure 4: Test accuracy versus total parameters (each point is one model).

Inference. Accuracy is not a function of parameter count. The overhead of all modules is at most 2.8 % of the model, yet the points separate into two clusters: the baseline-like models at 23.60–23.73 M parameters (91.2–92.3 %) and the three best models at 24.12–24.26 M (93.3–93.7 %). Models of equal size give different accuracy (Channel→Spatial P_3 and CBAM, both 23,730,215 parameters: 91.37 vs. 91.73 %), and a larger model can be worse (Channel→Spatial at

$P_3 + P_4$ with 24.26 M parameters: 91.19 % vs. 93.35 % for P_4 alone, 24.12 M). A large number of added parameters seems necessary for the gain seen here, but is not sufficient: the accuracy gain depends on *what* kind of attention is added and *where*.

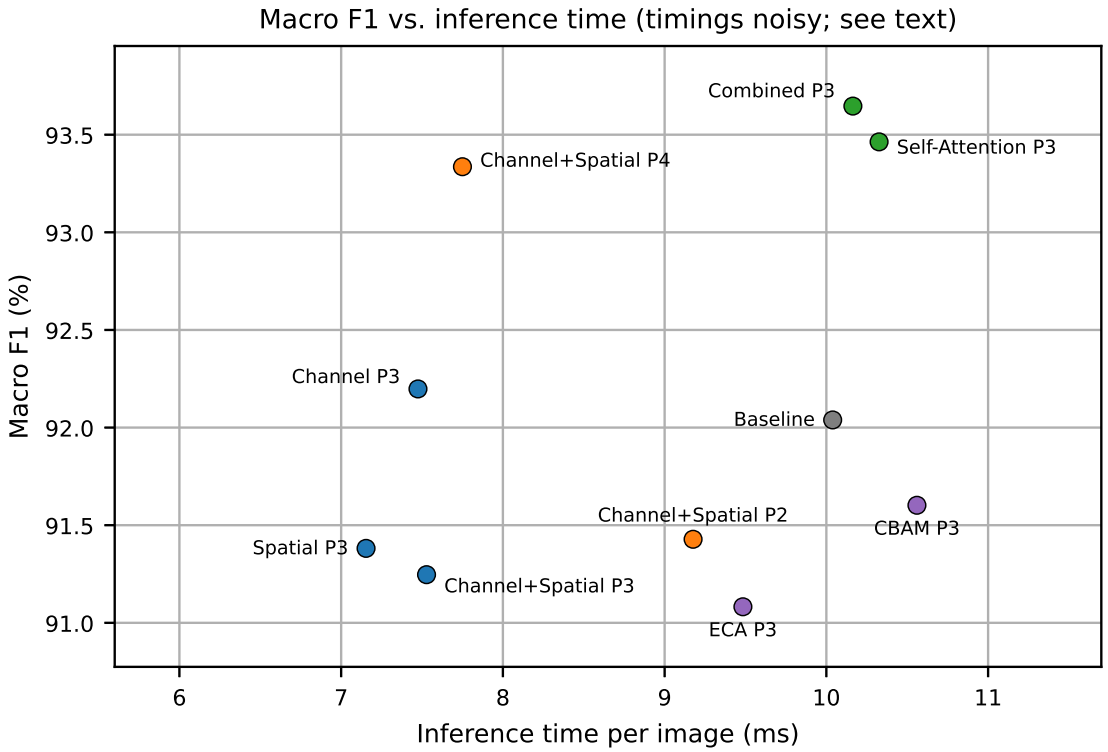


Figure 5: Macro F1 versus measured inference time per image.

Inference. No meaningful F1–time trade-off can be read from this plot. The points are scattered over 7.2–10.6 ms with no relation to F1 or to the cost of the module: the best three models take 7.75 ms (P_4), 10.16 ms and 10.33 ms, but the baseline takes 10.04 ms and the cheapest module (Spatial) 7.15 ms. As discussed in Section 10 the timing noise exceeds the true cost of the modules. The reliable conclusion is the analytical one: the extra computation of every module is small ($\lesssim$2.7 % of ResNet50's multiply-accumulates for self-attention, far less for the others), so the F1 gains of the best models do not come at a substantial latency cost. To quantify latency properly, repeat the timing over many passes with GPU synchronisation.

11. Training and Validation Graphs

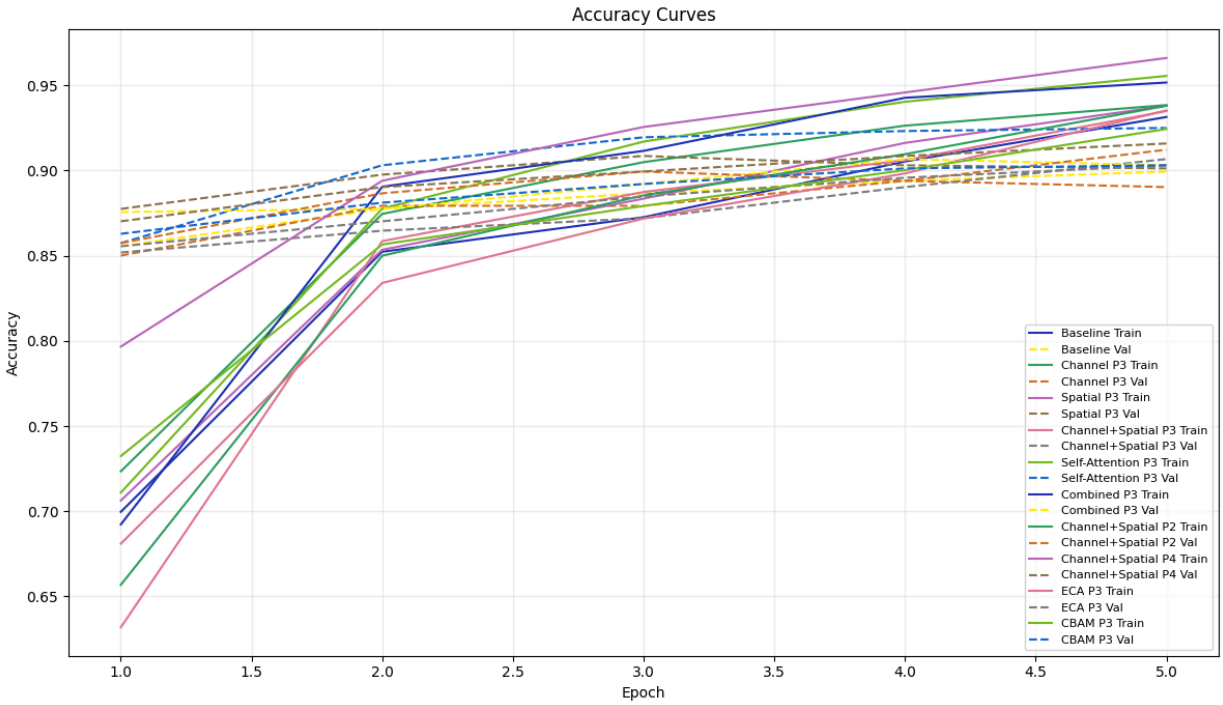


Figure 6: Training (solid) and validation (dashed) accuracy per epoch for all ten models (epochs 1–3: frozen backbone, epochs 4–5: fine-tuning). The default colour cycle repeats every ten curves, so individual curves are not unambiguously identifiable; the comparison between models relies on the test results of Section 9.

Inference. Training accuracy starts between about 0.63 and 0.80 in epoch 1 and rises steeply to 0.85–0.89 in epoch 2 and to roughly 0.93–0.97 in epoch 5. Validation accuracy starts higher (about 0.85–0.88) because the training metric is computed with augmentation and dropout active, and thereafter increases slowly to a narrow band of about 0.89–0.925. The train–validation gap therefore *opens* in the fine-tuning epochs (4–5) for most models, a sign of mild over-fitting once conv5_x is unfrozen; no curve shows instability or a collapse. After one or two epochs the validation accuracies of baseline and attention models are already similar, so the data do not show that attention accelerates convergence or reduces the train–validation gap; with only five epochs such differences could not be resolved anyway.

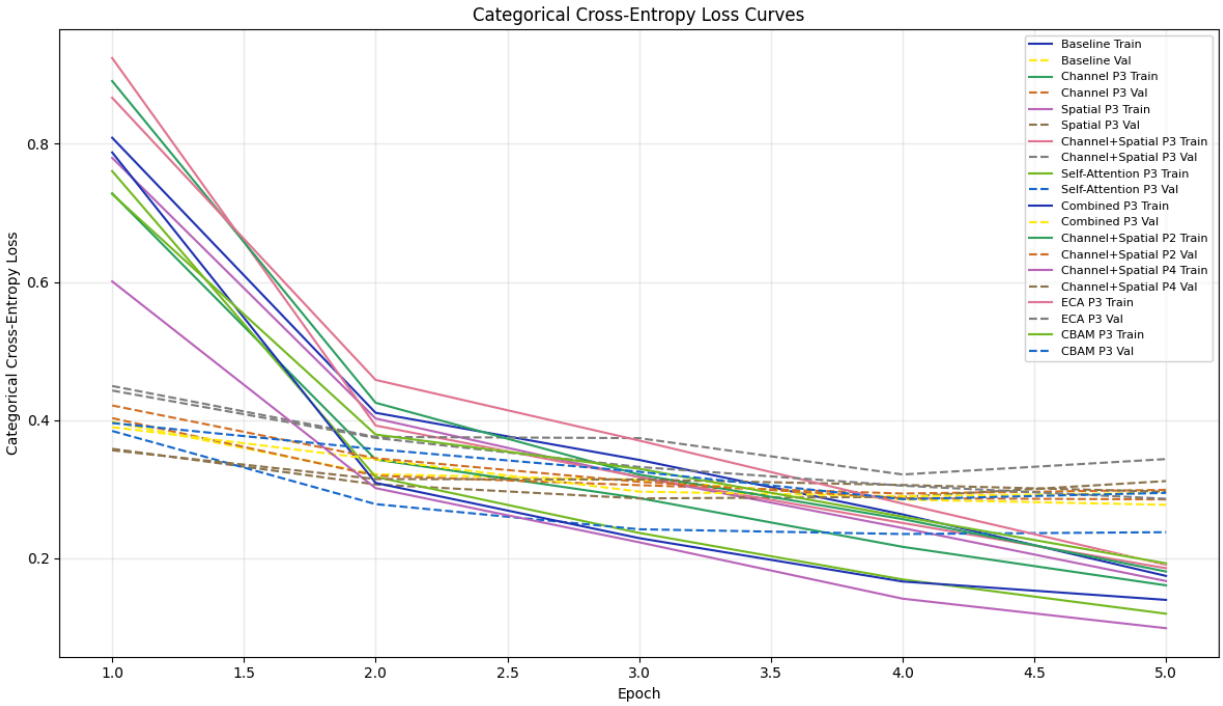
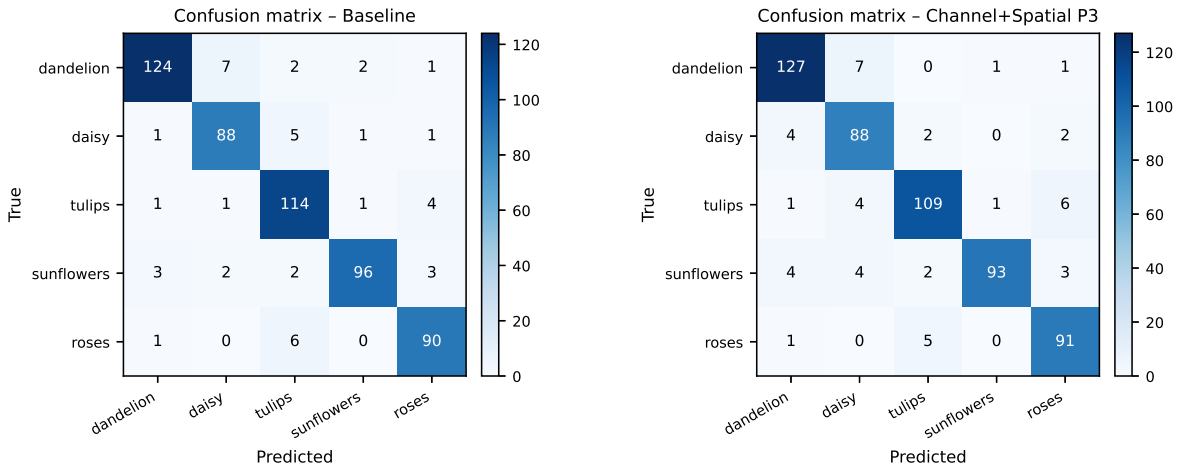


Figure 7: Training (solid) and validation (dashed) categorical cross-entropy for all ten models.

Inference. Training loss falls from about 0.6–0.9 in epoch 1 to about 0.10–0.19 in epoch 5. Validation loss decreases much less (from about 0.36–0.45 to 0.24–0.34) and is flat or slightly rising in the last epoch for several models, so by epochs 4–5 the training loss lies below the validation loss for all models and the gap widens during fine-tuning. This matches the accuracy curves: generalisation saturates quickly, and additional training would mostly lower the training loss. The attention models do not show unstable loss behaviour, but they also do not converge distinctly faster than the baseline.

12. Confusion Matrix Analysis



(a) Baseline ResNet50 (512/556 correct).

(b) Channel→Spatial at P_3 (508/556 correct).

Figure 8: Test-set confusion matrices (rows: true class, columns: predicted class).

Table 13: Class-wise precision (P), recall (R) and F1 in % on the test set. The Self-Attention confusion matrix was not stored in the notebook output, but its class-wise report was.

	Baseline			**Channel→Spatial P_3**			**Self-Attention P_3**		
Class	P	R	F1	P	R	F1	P	R	F1
dandelion	95.4	91.2	93.2	92.7	93.4	93.0	94.2	96.3	95.3
daisy	89.8	91.7	90.7	85.4	91.7	88.4	91.9	94.8	93.3
tulips	88.4	94.2	91.2	92.4	90.1	91.2	91.7	91.7	91.7
sunflowers	96.0	90.6	93.2	97.9	87.7	92.5	98.0	90.6	94.1
roses	90.9	92.8	91.8	88.3	93.8	91.0	91.9	93.8	92.9
Macro avg.	92.1	92.1	92.0	91.4	91.3	91.2	93.6	93.4	93.5

Inference.

- **Best-classified classes.** The baseline recognises tulips best (recall 94.2 %, 114/121) and sunflowers/dandelion have the highest F1 (93.2 %); the weakest F1 is daisy (90.7 %). With Channel→Spatial the best recall is for roses (93.8 %) and dandelion (93.4 %) and the lowest for sunflowers (87.7 %).
- **Frequent confusions.** In the baseline the largest off-diagonal entries are dandelion→daisy (7), roses→tulips (6) and daisy→tulips (5). Dandelion/daisy (small bright heads) and roses/tulips (similar petal colours and cup-like shapes) are plausible confusable pairs. The Channel→Spatial model keeps dandelion→daisy (7), has tulips→roses (6) and roses→tulips (5), and adds new errors: sunflowers→dandelion (4), sunflowers→daisy (4), daisy→dandelion (4) and tulips→daisy (4).
- **Which class benefits.** Channel→Spatial at P_3 improves dandelion (124→127 correct; the dandelion→tulips error disappears) and roses (+1), leaves daisy unchanged (88) and loses tulips (−5) and sunflowers (−3): net −4 images (91.37 % vs. 92.09 %). In contrast, Self-Attention at P_3 improves the F1 of *all five* classes (daisy +2.6 pp, dandelion +2.0, roses +1.0, sunflowers +0.9, tulips +0.5); the gain is largest for daisy and dandelion, whose recall rises from 91.7 % to 94.8 % and from 91.2 % to 96.3 %. The tulips recall is 2.5 pp lower (94.2 %→91.7 %) but the tulips precision is higher (88.4 %→91.7 %).
- **Distribution of the improvement.** The overall difference of Channel→Spatial is the net of gains and losses in different classes (an accuracy decline of 0.7 pp hides a +3 image gain for dandelion), whereas the Self-Attention improvement is spread over all classes, which supports it being a general effect rather than one class. The numbers are small (a handful of images), so these per-class statements are indicative.

13. Grad-CAM Analysis

Grad-CAM gives a class-specific map of the regions that contribute to a prediction. For feature map A^k of a convolutional layer and class score y^c, $\alpha_k^c = \frac{1}{HW}\sum_{i,j}\frac{\partial y^c}{\partial A_{ij}^k}$ and $L^c_{\text{Grad-CAM}} = \text{ReLU}\big(\sum_k \alpha_k^c A^k\big)$, resized to the image resolution and overlaid on the input. It is a *post-hoc* explanation computed from gradients after the prediction, whereas an internal attention map is computed by the module during the forward pass and changes the features (Section 14).

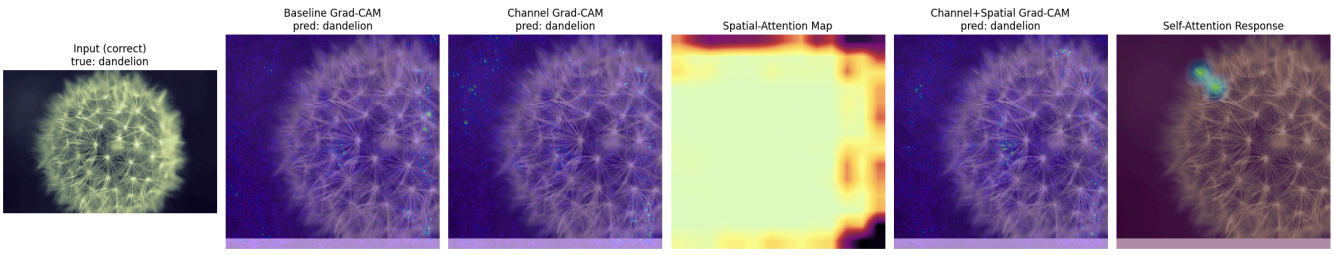


Figure 9: Correctly classified test image (true class: dandelion). Columns: input; Grad-CAM-style maps of the baseline, of channel attention (P_3) and of Channel+Spatial P_3 (columns 2, 3, 5); spatial-attention map of Spatial P_3 (column 4); mean self-attention response of Self-Attention P_3.

Visible evidence. All three Grad-CAM-style overlays (baseline, channel, channel+spatial) predict *dandelion* and look almost identical: a diffuse, fine-grained speckle spread over the whole frame, including the dark background on the left, with a few brighter specks on the right-hand part of the seed head. None produces a compact region

over the flower centre, and the attention models do not show visibly weaker activation on the background than the baseline. The learned spatial-attention map is close to uniform and high over most of the frame and drops only at the right and bottom borders, i.e. it does not separate flower from background. The mean self-attention response is low almost everywhere and concentrates in one small spot on the upper-left part of the seed head, which suggests that a few positions receive most of the attention from the other positions.

Interpretation and caveats. (i) The speckled, pixel-level texture of the Grad-CAM-style maps is typical of an *input-gradient* saliency map rather than a smooth heat-map at the 14×14 or 7×7 feature resolution; the notebook's `gradcam` function falls back to an input-gradient map when the intermediate sub-model cannot be built, and this is consistent with what is seen. These maps should therefore be called gradient-based saliency overlays, and a proper layer-level Grad-CAM (e.g. on `conv5_block3_out`) should be generated before drawing localisation conclusions. (ii) In the overlay panels the picture is drawn on the 224×224 extent of the map, so it appears stretched and a translucent band is visible at the lower edge; alignment of a hot spot with a flower part is therefore approximate. (iii) Only the images of Figures 9 and 13 were stored, both correctly classified dandelions, so nothing can be concluded about other classes or about errors.

14. Internal Attention-Map Analysis

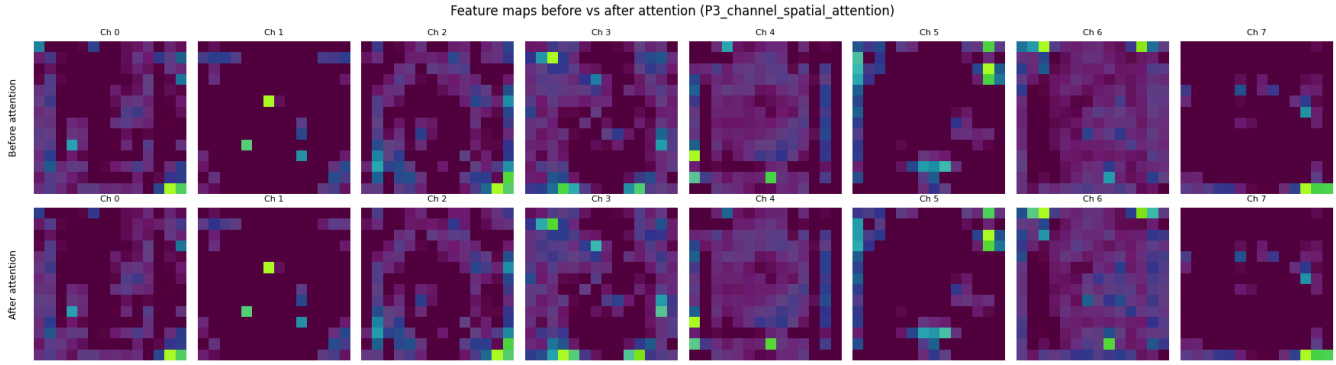


Figure 10: First eight channels of the $14 \times 14 \times 1024$ feature map before (top) and after (bottom) the Channel→Spatial block at P_3 for the first test image. Each panel is min–max normalised.

Inference. The maps before and after attention are visually almost identical: the same sparse peaks (e.g. channel 1, channel 7) and the same regions of weak response remain. The block therefore acts as a mild re-calibration of the features rather than a hard selection. Because every panel is min–max normalised, a constant gain on a channel would not be visible, so the figure shows only the *relative spatial* redistribution of activation, which is small – consistent with the almost flat spatial map of Figure 9.

Channel attention.

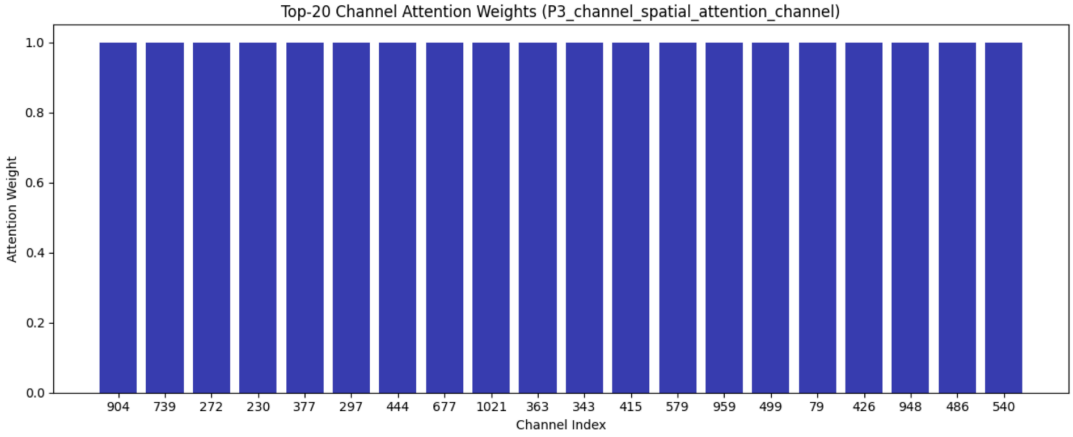


Figure 11: Top-K channel-attention weights for Channel→Spatial at P_3.

Status. The channel-attention weights are an *internal* quantity computed in the forward pass (the `last_attention` attribute of the channel module); the figure and the comments on how many channels receive strong weights, and whether the weights change between input images, cannot be given because the output was not stored. The only

available indirect evidence is the before/after feature map above, in which the channel re-weighting visibly changes little.

Spatial attention. Obtained. The spatial-attention map of Spatial P_3 (fourth panel of Figures 9 and 13) is 14×14 and is shown resized to 224×224. It is high and nearly uniform in the interior of the image and lower along the borders (right and lower edges in Figure 9, left and right edges in Figure 13). It therefore has no preference for the flower region: the highest values cover the flower, the surrounding background and the leaves alike. This is consistent with the module's small effect on accuracy (91.55 % vs. 92.09 %) and with its 98 parameters. The standalone "Input | Attention map | Overlay" plot of the notebook was not stored.

Self-attention. Obtained. The self-attention matrix is 196×196 at P_3. Following the notebook, its mean over the query positions is plotted, i.e. how strongly each position is attended to on average (sixth panel of Figures 9 and 13). In both images the response is low almost everywhere and concentrated in one compact spot (upper-left part of the seed head in Figure 9, lower-central part in Figure 13). Thus long-range interaction is dominated by a few key positions to which most other positions attend, rather than being spread uniformly over the flower. The average over queries hides query-specific patterns; maps for selected query locations would show better how distant regions interact. The spot position also differs between the two images, i.e. the pattern is image-dependent.

Attention at different positions.

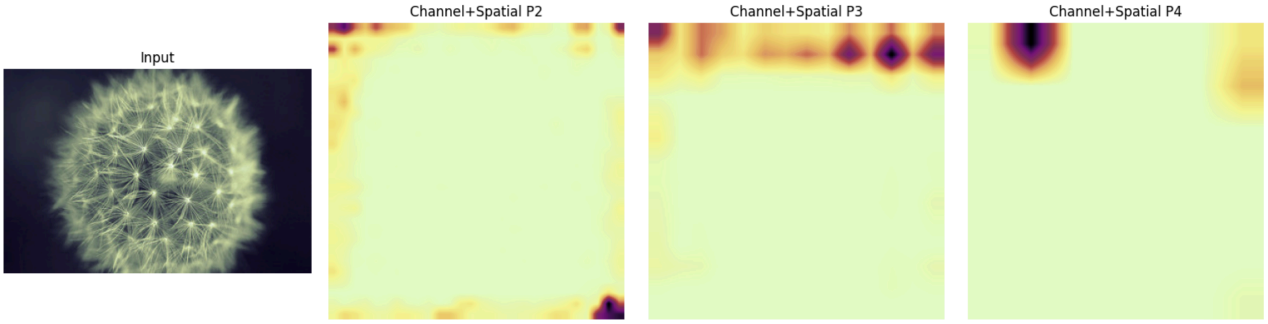


Figure 12: Spatial-attention maps of Channel→Spatial at P_2, P_3 and P_4 for the same test image.

Status. The three models are trained and evaluated (Section 8), and the maps would be 28×28, 14×14 and 7×7, but the Manual requires the observations to rest on the student's own visualisations and the image was not stored, so no statement about how the maps change with depth is made here.

15. Qualitative Comparison

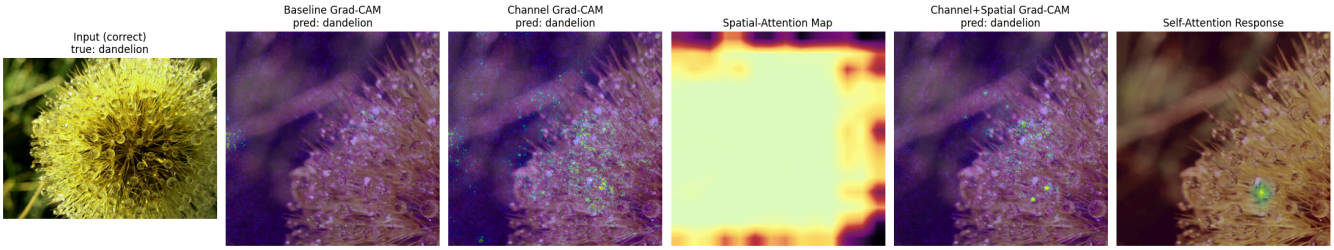


Figure 13: Second stored test image (a dandelion with water droplets, correctly classified; the panel title of the notebook reads "correct"). Same panel order as Figure 9.

Visible evidence. All models predict dandelion. The baseline overlay is weak and diffuse, with bright specks only in the right half of the seed head. The channel and channel+spatial overlays contain denser, brighter specks in the central and right part of the seed head and fewer on the dark upper-left background, i.e. a modest shift of the response from the background to the flower; their maps differ little from each other. The spatial-attention map is again flat and high over most of the frame, and the self-attention response is a single spot in the lower-central part of the seed head.

Discussion against the criteria. *Foreground localisation / background suppression:* slight in image 2, absent in image 1. *Compactness vs. coverage:* the gradient-based maps are diffuse and cover the whole frame; the self-attention response is very compact; the spatial map covers almost everything. *Flower centre vs. petals:* the speckles in image 2 lie on the dense central region of the seed head, but the resolution of these maps does not allow a petal-level statement. *Shallow vs. deep attention:* only P_3 maps were stored. *Consistency:* the spatial gate is near-uniform

in both images, whereas the self-attention spot and the Grad-CAM-style speckle patterns change from image to image. *Correct vs. incorrect images:* the requested set of three correct and two incorrect images was generated by the notebook, but only these two correct examples were stored; no comparison with misclassified samples can be reported. With two images of the same class the qualitative evidence is anecdotal and consistent with, but not a confirmation of, the quantitative result that the accuracy gain of the best models is small and not clearly linked to sharper localisation.

16. Additional Attention-Mechanism Comparison

Two additional mechanisms, ECA and CBAM, were compared with the laboratory modules on the same ResNet50 at P_3 (SE is identical to the channel attention of Section 4 and is reported as "Channel / SE"; coordinate attention was not implemented).

Table 14: Comparison of attention mechanisms at P_3 (macro-averaged test percentages).

Attention	Accuracy	Precision	Recall	Macro F1	Parameters	Inference Time
No Attention	92.09	92.09	92.08	92.04	23,597,957	10.04 ms
Channel / SE (r=16)	92.27	92.52	92.10	92.20	23,730,117	7.47 ms
Spatial Attention	91.55	91.56	91.42	91.38	23,598,055	7.15 ms
ECA	91.19	91.05	91.14	91.08	23,597,960	9.48 ms
CBAM	91.73	92.00	91.40	91.60	23,730,215	10.56 ms
Self-Attention	93.53	93.56	93.45	93.46	24,122,246	10.33 ms
Additional: Combined (C+S+SA)	93.71	93.86	93.59	93.65	24,254,504	10.16 ms

All models in this table use the same ResNet50 backbone, position P_3, split and training protocol. All metrics are macro-averaged test percentages. SE is mathematically the same module as the channel attention of Section 4 (global average pooling, bottleneck $r = 16$, sigmoid), so it was trained once and is reported as "Channel / SE". The notebook's ECA is a 1-D convolution with $k = 3$ over the pooled channel descriptor; its CBAM uses a shared MLP on both average- and max-pooled channel descriptors, followed by a 7×7 spatial gate. The inference time is the notebook's mean per-image time (see Section 10 for caveats). Coordinate attention was not implemented.

1. **Accuracy.** It ranges from 91.19 % (ECA) to 93.71 % (Combined). Self-attention and Combined are best (≥93.5 %); the baseline, Channel/SE, Spatial and CBAM lie within 91.5–92.3 %; ECA is lowest.
2. **Macro F1.** It follows accuracy closely (Combined 93.65, Self-Attention 93.46, Channel/SE 92.20, Baseline 92.04, CBAM 91.60, Spatial 91.38, ECA 91.08); because the classes are fairly balanced, accuracy and macro F1 never differ by more than 0.2 pp.
3. **Parameters.** Channel/SE adds 132,160; Spatial 98; ECA 3; CBAM 132,258; Self-attention 524,289; Combined 656,547 (at most +2.8 % of the 23.6M-parameter baseline).
4. **Inference time.** The measured values (7.15–10.56 ms/image) are not ordered by module cost – e.g. the baseline (10.04) is slower than Channel/SE (7.48), and CBAM and Channel→Spatial have exactly the same parameter count but 10.56 vs. 7.53 ms. The measurement noise is larger than the true cost of the modules, so no timing conclusion is drawn (Section 10).
5. **Convergence.** In the five-epoch curves (Section 11) all models reach a validation accuracy of about 0.85–0.88 after the first epoch, regardless of the attention type; no attention model converged visibly faster or more smoothly.
6. **Feature localisation.** The saved visualisations show no clear sharpening towards the flower (Sections 13 and 15): the Grad-CAM-style maps of the baseline and the attention models look alike, and the learned spatial gate is almost uniform.
7. **Does more complexity always help?** No. Self-attention (+524k) helps, but CBAM and Channel→Spatial (+132k) do not, and Channel→Spatial at $P_3 + P_4$ (+659k) is worse than P_4 alone (+527k).
8. **Lightweight vs. heavy.** Channel/SE (+132k, 92.27 %) is as good as the baseline and better than the larger CBAM (91.73 %) at equal size; ECA (3 parameters) is 1.08 pp below Channel/SE. The lightweight modules do not reach the accuracy of self-attention, which costs about four times more parameters than SE.

17. Discussion

Quantitative (33.1). Baseline: 92.09 % accuracy / 92.09 % precision / 92.08 % recall / 92.04 % macro F1. Best: Combined P_3 (93.71 / 93.86 / 93.59 / 93.65) and Self-Attention P_3 (93.53 / 93.56 / 93.45 / 93.46), followed by Channel→Spatial at P_4 (93.35 / 93.41 / 93.34 / 93.34). Gains are +1.3 to +1.6 pp (7–9 images of 556); every other configuration lies within ±1 pp of the baseline.

Complexity (33.2). Extra parameters range from 3 (ECA) to 656,547 (Combined), i.e. at most 2.8 % of the model. The analytical extra computation is $\lesssim$2.7 % of ResNet50's MACs; the measured inference times (7.15–10.56 ms/image) are dominated by noise and cannot rank the modules.

Qualitative (33.3). The attention modules' own maps are weakly selective (an almost flat spatial gate; a single-spot mean self-attention response). The gradient-based overlays of the baseline and attention models are similar, with a slight shift toward the seed head for the channel models in one of two images. Visual evidence for background suppression is weak, and based on two images only. Quantitative and qualitative observations are thus consistent in that the benefit of attention in this setting is modest.

- **Why performance changed.** Self-attention at P_3 (+1.44 pp) and the Channel→Spatial block at P_4 (+1.26 pp) are the configurations that improved on the baseline. Both are "safe" ways of adding a module to a pretrained network: self-attention is residual with γ starting at 0 (it begins as an identity), and the P_4 module feeds only newly trained layers. The sigmoid-gated modules at P_2/P_3 rescale features that the pretrained downstream stages expect unchanged, which can offset any benefit within five epochs. This is a hypothesis consistent with all results, not something tested by a separate ablation.
- **Location and resolution.** The same module gave 91.55 %, 91.37 % and 93.35 % at P_2, P_3 and P_4. Channel attention is more useful on the 2048 high-level channels of conv5_x than on 512/1024 mid-level channels, and its cost grows with the number of channels. The spatial component has little to select on a 7×7 map, and the observed spatial gate at P_3 was nearly flat.
- **Complexity versus gain.** The best gains cost 2.2–2.8 % extra parameters and at most a few percent of extra computation. The inexpensive modules (Spatial, ECA) did not help, and stacking modules ($P_3 + P_4$) increased cost and *reduced* accuracy; bigger or more attention is not automatically better.
- **Do the maps correspond to the flower?** Only weakly. The learned spatial gate and the gradient-based overlays show little preference for the flower over the background, and the mean self-attention response concentrates on a few positions. Hence no claim of better background suppression is supported; it would need layer-level Grad-CAM on many images.
- **Quantitative versus qualitative.** Both indicate modest effects: small (one-standard-error scale) accuracy differences and visually similar maps for the baseline and attention models.
- **Limitations.** One run per configuration (seed 42), a short 3+2 epoch schedule, 556 test images (1 image = 0.18 pp), noisy inference-time measurements, Grad-CAM-style maps that appear to be input-gradient saliency, only two stored qualitative images (both correct), and results not stored in the notebook file for the Spatial→Channel order, the $3 \times 3/5 \times 5$ spatial kernels, the channel weights, the $P_2/P_3/P_4$ maps and the $P_2{+}P_3{+}P_4$ model.

Table 15: Status of the 14 required visualisations. "Not stored" means the plotting code exists in the notebook, but its image was not saved in the notebook file.

No.	Visualisation	Status
1	Sample images of all five classes	Figure 1
2	Training/validation accuracy curves	Figure 6
3	Training/validation loss curves	Figure 7
4	Confusion matrix (baseline and attention model)	Figure 8
5	Feature map before attention	Figure 10 (top row)
6	Channel-attention weights	Figure 11
7	Spatial-attention heat-map	Figures 9, 13 (4th panel)
8	Spatial-attention overlay (Input | map | overlay)	stored
9	Self-attention response	Figures 9, 13 (6th panel)
10	Feature map after attention	Figure 10 (bottom row)
11	Baseline Grad-CAM	Figures 9, 13
12	Grad-CAM of attention-based models	Figures 9, 13
13	Attention maps at P_2, P_3, P_4	Figure 12
14	Graphical comparison of attention mechanisms	Figures 2–5

Answers to the Laboratory Questions

The answers below retain the reported experimental values and conclusions while using revised wording.

1. What is the purpose of residual connections in ResNet50?

Answer: A residual block adds its input to the block's learned transformation, giving $y = F(x) + x$. This direct path helps gradients travel through a deep network and lets a block learn a refinement rather than having to recreate the entire representation.

2. What does conv2_x represent compared with conv5_x?

Answer: The conv2_x stage produces relatively detailed $56 \times 56 \times 256$ feature maps, useful for edges, colours and textures. By conv5_x, the maps are $7 \times 7 \times 2048$ and encode more abstract, class-relevant patterns with a larger receptive field.

3. Why does the spatial resolution decrease with network depth?

Answer: Strided convolutions and pooling progressively downsample the feature maps ($224 \rightarrow 112 \rightarrow 56 \rightarrow 28 \rightarrow 14 \rightarrow 7$). This reduces computation while allowing deeper units to use a wider receptive field and capture broader context.

4. Why does the number of feature channels increase with network depth?

Answer: As the feature maps become smaller, later stages increase their channel count from 64 to 256, 512, 1024 and 2048. The extra channels let the network represent a wider range of increasingly abstract patterns.

5. What is meant by low-level, intermediate, and high-level features?

Answer: Low-level features include edges, colours and simple textures. Intermediate features represent structures such as petals and flower parts. High-level features encode more complete, class-discriminative information, typically found in the deeper conv5_x stage.

6. What is the purpose of an attention mechanism?

Answer: Attention reweights selected channels, spatial locations or relationships so that useful evidence has greater influence and less useful information is reduced. In these experiments, the clearest gains beyond roughly one standard error came from P3 self-attention and Channel→Spatial attention at P4.

7. What does channel attention attempt to learn?

Answer: It learns a separate sigmoid weight for each channel using a globally pooled feature descriptor. These weights indicate which feature types are most useful to the prediction.

8. What does spatial attention attempt to learn?

Answer: Spatial attention creates a weight map over image locations. It uses channel-wise average and maximum summaries followed by a convolution and sigmoid to estimate where informative evidence may occur.

9. How is channel attention different from spatial attention?

Answer: Channel attention produces a $1 \times 1 \times C$ vector that weights feature types across the whole map; spatial attention produces an $H \times W \times 1$ mask shared by the channels. At P3, the reported additions were 132,160 parameters for channel attention and 98 for spatial attention.

10. Why can channel and spatial attention be used sequentially?

Answer: The two modules operate along different dimensions of the same feature tensor, so one can select feature types and the next can refine their locations. However, combining them did not automatically improve performance: Channel→Spatial at P3 scored 91.37%, below channel attention (92.27%) and spatial attention (91.55%).

11. Why might channel attention behave differently at conv3_x and conv5_x?

Answer: The 512 channels at conv3_x generally represent reusable mid-level patterns, whereas the 2048 channels at conv5_x are more semantically specific. Reweighting those deeper channels can therefore affect class evidence differently. The measured Channel→Spatial results were 91.55% at P2 and 93.35% at P4.

12. Why can spatial attention be useful at an intermediate ResNet stage?

Answer: Intermediate maps such as 28×28 or 14×14 retain enough detail to distinguish object parts while containing meaningful features. In this run, though, Spatial P3 reached 91.55% and its learned map was almost uniform, so the expected benefit was not observed.

13. Why is self-attention computationally expensive when the feature map has high spatial resolution?

Answer: Self-attention forms an $N \times N$ matrix for $N = H \times W$ positions, so its time and memory requirements grow quadratically with N. The matrix has 38,416 entries at P3, 614,656 at P2 and 9,834,496 at P1.

14. What happens to the size of the self-attention matrix when the spatial resolution increases?

Answer: The matrix size scales as $(HW)^2$. Doubling both height and width makes four times as many positions and therefore sixteen times as many pairwise entries. The listed sizes from P4 through P1 are 2,401, 38,416, 614,656 and 9,834,496 entries.

15. How does attention performance vary when the insertion position is changed?

Answer: For Channel→Spatial attention, accuracy was 91.55% at P2, 91.37% at P3 and 93.35% at P4, compared with 92.09% for the baseline. P4 performed best, although each position was tested only once, so the difference is evidence rather than definitive proof.

16. How does attention at P2 differ visually from attention at P4?

Answer: The saved report does not include the P2/P3/P4 attention-map comparison, so a visual difference cannot be confirmed from the available outputs. Their resolutions are known, however: P2 is 28 × 28 and P4 is 7 × 7, making P4 much coarser.

17. Does applying attention at multiple stages always improve classification performance?

Answer: No. Using Channel→Spatial at both P3 and P4 achieved 91.19%, lower than P4 alone at 93.35% and P3 alone at 91.37%. It also added about 132,000 parameters beyond the P4 model.

18. Does Channel→Spatial attention behave differently from Spatial→Channel attention?

Answer: The recorded results cover Channel→Spatial only (91.37% accuracy and 91.25% F1 at P3). The Spatial→Channel output was not saved, so this experiment cannot establish an empirical difference. The two orders can differ because each second module receives features already reweighted by the first.

19. How does the spatial-attention kernel size influence the attention map?

Answer: A larger kernel uses a wider neighbourhood when estimating each location's weight. For kernel sizes 3, 5 and 7, the spatial convolution sees 9, 25 and 49 positions respectively. The saved 7 × 7 result was smooth and nearly flat; results for the smaller kernels were not saved.

20. How does SE attention differ from ECA attention?

Answer: SE pools the feature map and uses a two-layer bottleneck MLP to learn channel weights, allowing broad interactions between channels. ECA instead applies a short 1-D convolution to neighbouring entries of the pooled descriptor and avoids channel reduction. At P3, SE/channel attention scored 92.27%, while ECA scored 91.19%.

21. How does CBAM combine channel and spatial attention?

Answer: CBAM first computes channel weights using average- and max-pooled descriptors passed through a shared MLP, then estimates a spatial mask from channel-wise average and maximum maps. In this run, CBAM P3 had the same added parameter count as Channel→Spatial P3 (132,258) and scored 91.73% versus 91.37%.

22. How does self-attention differ from channel attention?

Answer: Self-attention calculates relationships between spatial positions and uses query, key and value projections to combine information across positions, with a residual connection. Channel attention instead assigns weights to channels from a global pooled descriptor. Self-attention was the only individual attention type to show a clear baseline improvement in this experiment.

23. Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?

Answer: No. The Combined module had the most added parameters (+656,547) and was only one test image better than self-attention. The P3+P4 model added +658,820 parameters but reached just 91.19%, while Channel→Spatial at P4 achieved 93.35%. Parameter count alone did not predict accuracy.

24. What is the difference between an internal attention map and Grad-CAM?

Answer: An internal attention map is learned as part of the forward pass and directly reweights features used for prediction. Grad-CAM is calculated after a prediction by examining gradients of a target class score with respect to convolutional features; it explains a prediction but does not change it.

25. Why is Grad-CAM called a post-hoc explanation technique?

Answer: It is generated after the model has produced a prediction, using gradients to estimate which feature locations contributed to the selected class. It is not part of the model's training process or the prediction computation itself.

26. Does the attention model suppress irrelevant background regions?

Answer: The stored visualisations do not provide convincing evidence of background suppression. The attention overlays look broadly similar to the baseline and the learned spatial gate is almost uniform, with only a small increase around the seed head in one image. The comparison covers only two images.

27. Are the attention maps of correctly classified and incorrectly classified images different?

Answer: The available figures cannot answer this reliably. Although the notebook selected correct and incorrect test examples, the saved figures show only correctly classified dandelions. Channel→Spatial P3 misclassified 48 of the 556 test images.

28. Which flower classes benefit most from attention?

Answer: For P3 self-attention, F1 improved for all five classes, with the largest gains for daisy (+2.6 percentage points) and dandelion (+2.0 points). Channel→Spatial P3 helped dandelion and roses, but had fewer correct predictions for tulips (−5) and sunflowers (−3).

29. What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?

Answer: A confusion matrix shows which categories are mistaken for one another and how often, allowing class-level precision and recall to be calculated. For example, the baseline confused dandelions with daisies seven times, but daisies with dandelions only once. Models with identical accuracy can still make different errors.

30. Why is macro F1 useful in addition to accuracy?

Answer: Macro F1 calculates F1 for each class and gives every class equal weight, so weaker classes are not hidden by stronger ones. The dataset is reasonably balanced and the reported macro F1 values were only 0.05–0.2 points below accuracy, but the metric still checks that gains are not driven by neglecting a class.

31. What is meant by the accuracy–complexity trade-off?

Answer: It describes the balance between predictive performance and the extra parameters, memory and computation needed to achieve it. Self-attention added about 2.2% of parameters for a 1.44-point gain, and Channel→Spatial at P4 added about 2.2% for a 1.26-point gain. Smaller additions such as ECA did not improve accuracy here.

32. What is meant by the F1–inference-time trade-off?

Answer: This compares a model's F1 score with the latency added by its attention module. The measured inference times, about 7.2–10.6 ms per image, were too noisy to support a reliable ranking. The analytical estimate suggests even the heaviest module adds no more than roughly 2.7% MACs.

33. Why must every ablation study use the same dataset split?

Answer: Using the same training, validation and test images ensures that performance differences are not caused by different sample difficulty. With 556 test images, sampling variation alone is around the scale of the observed differences. This study reused one stratified split generated with seed 42.

34. Why must the same training protocol be maintained when comparing different attention mechanisms?

Answer: If optimisers, learning rates, epochs, augmentation or batch size differ, the results cannot be attributed solely to the attention design. Here, all models used Adam at 10■³ then 10■■, three frozen-backbone epochs plus two fine-tuning epochs, batch size 32, dropout 0.3, the same augmentation and seed 42. Variation from using only one seed remains a limitation.

35. What conclusions can be drawn from the Grad-CAM comparison?

Answer: The baseline and attention models predicted the same class in the two saved examples and produced broadly similar gradient-based maps. Channel-based models showed a slightly stronger response around the seed head in one image, but overall background suppression was weak and spatial gates were nearly flat. These limited visualisations do not show that the accuracy gains resulted from better localisation.

18. Conclusion

This experiment compared channel, spatial, Channel→Spatial, self-attention, ECA, CBAM and combined attention at P3, along with Channel→Spatial attention at P2, P3, P4 and P3+P4, using the same pretrained ResNet50, tf_flowers split and training protocol. The baseline achieved 92.09% test accuracy and 92.04% macro F1. P3 self-attention reached 93.53%, the combined model reached 93.71%, and Channel→Spatial at P4 reached 93.35%; these were the only configurations whose gains exceeded roughly one standard error. Channel, spatial, ECA, CBAM and Channel→Spatial at P2/P3 remained within about one percentage point of the baseline. The insertion position mattered more than simply increasing the number of parameters: P4 outperformed P2 and P3, while placing attention at both P3 and P4 reduced accuracy compared with a single location. Parameter overhead was modest (up to about 2.8%), but measured inference times were too noisy to rank the variants. The internal maps and gradient-based overlays showed only weak selectivity for the flower, so improved background localisation cannot be claimed. These findings should be interpreted cautiously because each configuration used one seed, training lasted only five epochs, the test set contained 556 images, and several planned visualisations were not saved.

The results suggest that attention can help a pretrained ResNet50, but its effect depends on the module and insertion point. Further runs with multiple seeds, longer training and more saved qualitative examples would be needed to confirm the observed differences.